

EduGenie: Google Gemini Powered Learning Assistant

4. Project Planning Phase

The source describes EduGenie development through milestones and activities. This document organizes those documented milestones into a practical project-planning record without inventing dates or undocumented team assignments.

4.1 Planning Approach

The documented workflow is divided into four milestones: model selection and architecture, core functionalities development, frontend development, and deployment.

The repository structure contains eight project-stage folders, including Brainstorming & Ideation, Requirement Analysis, Project Design Phase, Project Planning Phase, Project Development Phase, Project Testing, Project Documentation, and Project Demonstration.

4.2 Milestone 1 — Model Selection and Architecture

Select Gemini 1.5 Pro for Q&A, summarization, quiz generation, and learning paths.

Select LaMini-Flan-T5-783M for concept explanation.

Establish the modular folder architecture and identify the frontend, backend, and dependency files.

4.3 Milestone 2 — Core Functionalities Development

Implement the explanation module.

Implement QnA.

Implement quiz generation and JSON handling.

Implement summarization.

Implement personalized learning recommendations.

Connect the modules to FastAPI REST endpoints.

4.4 Milestone 3 — Frontend Development

Build index.html with the documented task dropdown, text area, submit button, and result container.

Build style.css with responsive design and styled controls.

Integrate the form with the FastAPI backend through POST requests.

4.5 Milestone 4 — Deployment and Validation

Run the application locally with `uvicorn main:app --reload`.

Open `http://127.0.0.1:8000`.

Validate question answering, explanations, quiz generation, summarization, and personalized learning plans.

4.6 Planning Dependencies

Python 3.10+ must be available.

FastAPI, Uvicorn, and Jinja2 are required.

A Google Gemini API key is required for Gemini-based features.

4.7 Planning Outcome

The planning structure provides a traceable path from idea to requirements, architecture, implementation, frontend integration, testing, documentation, and demonstration.

Conclusion

The project plan is milestone-driven and feature-oriented. It keeps the five user-facing AI capabilities aligned with the architecture, API endpoints, frontend workflow, and documented validation activities.

Source basis: EduGenie project document supplied with this request and the repository structure shown in the supplied reference image. Where the source did not specify an implementation detail, this documentation preserves the documented scope rather than inventing project facts.